

Ideal Number of Extrusions for Uniformity in Liposome Nanoparticle Synthesis

Group B11

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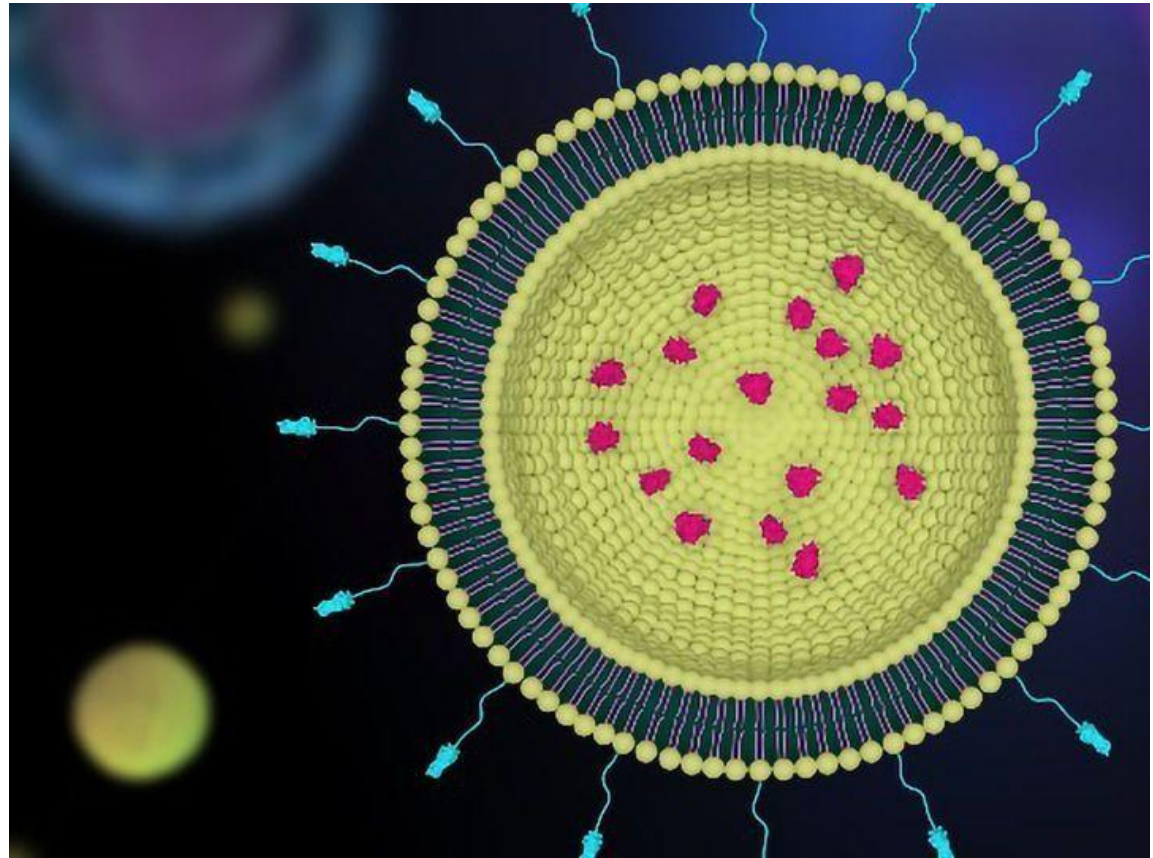
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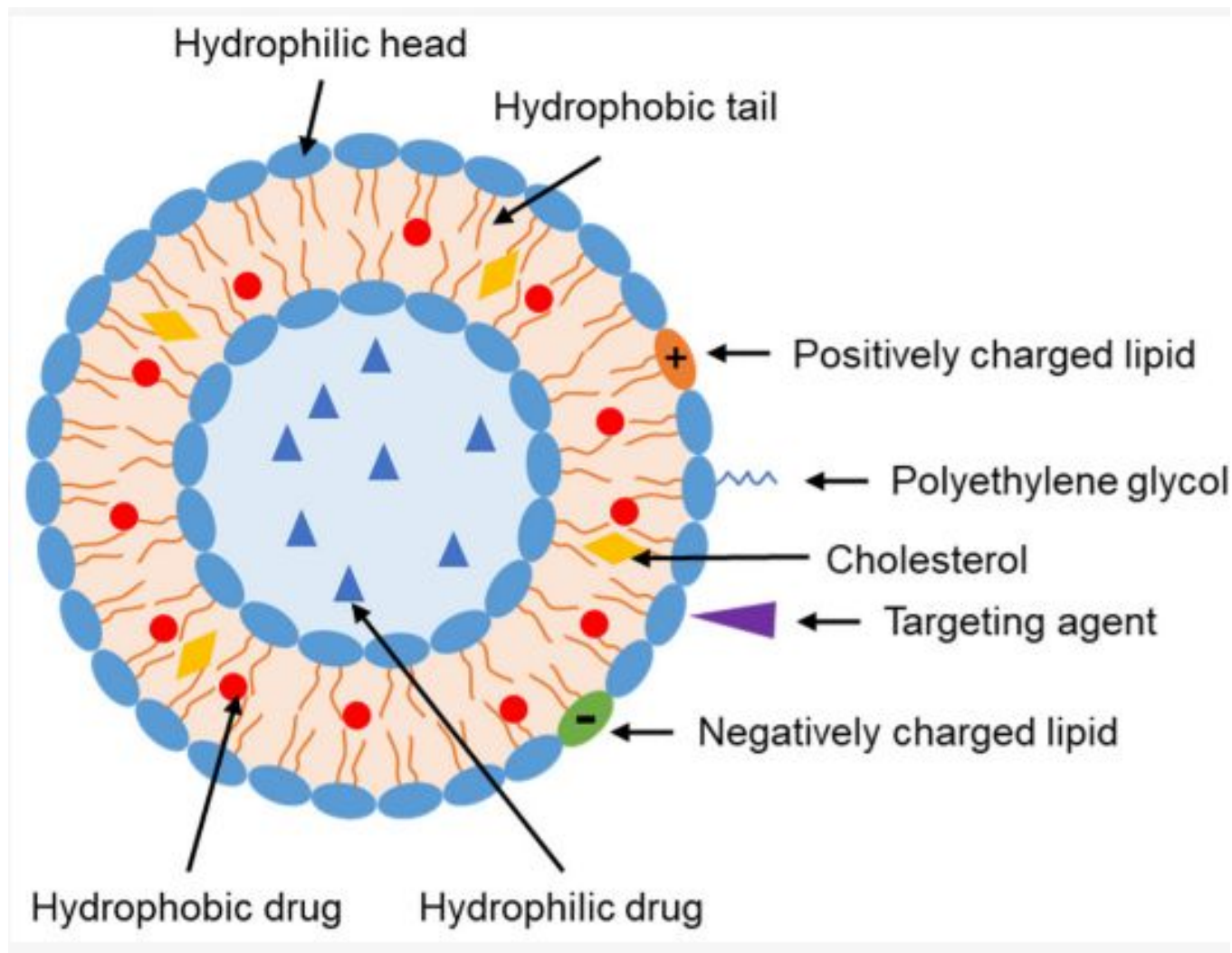


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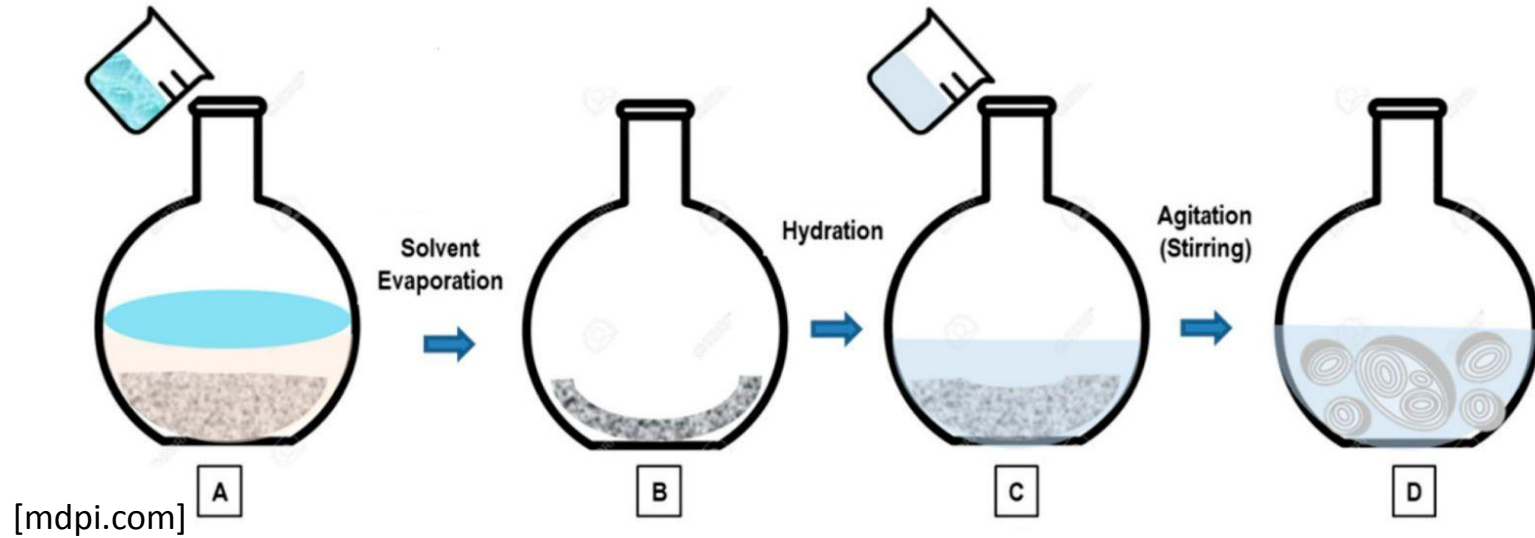
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[technologynetworks.com]

Due to their unique properties, liposome nanoparticles are commonly used in medicine and drug delivery.

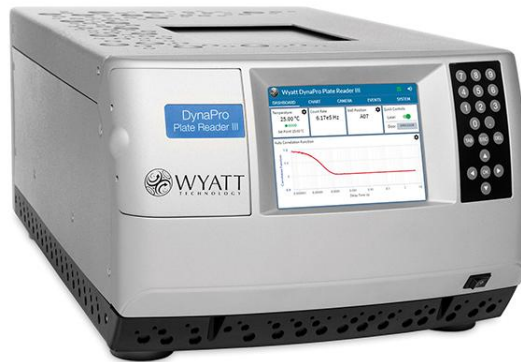


Liposome nanoparticles are formed through lipid film hydration and

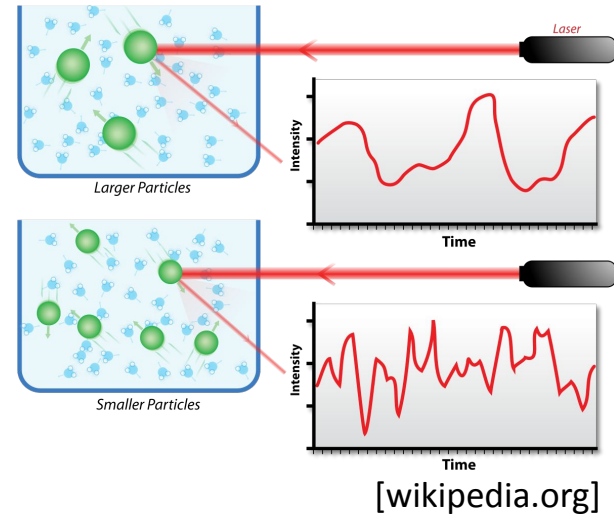


[avantilipids.com]

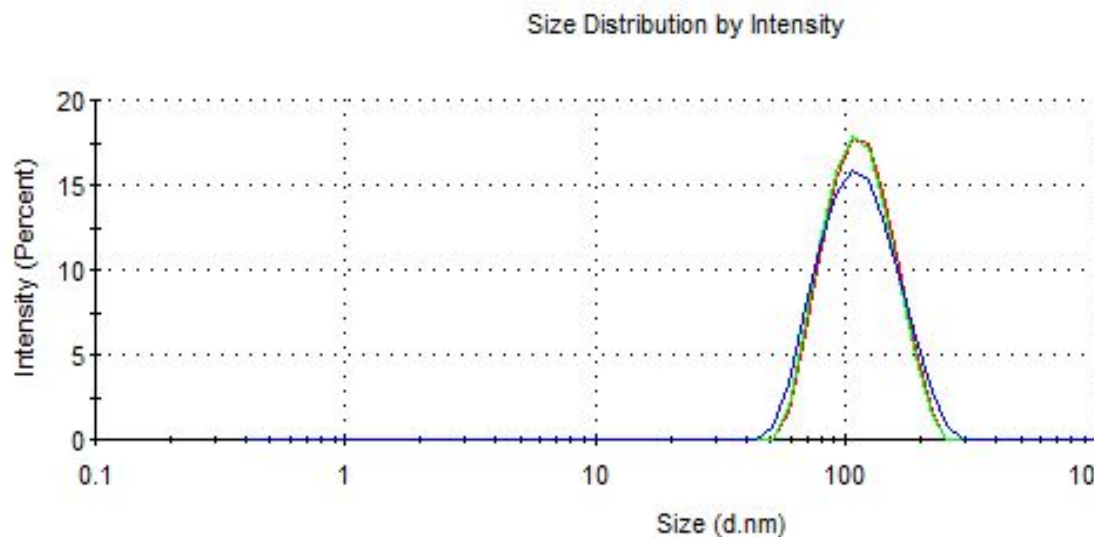
Dynamic Light Scattering (DLS) is used to determine the average particle diameter and Polydispersity Index (PDI).



[md-scientific.dk]

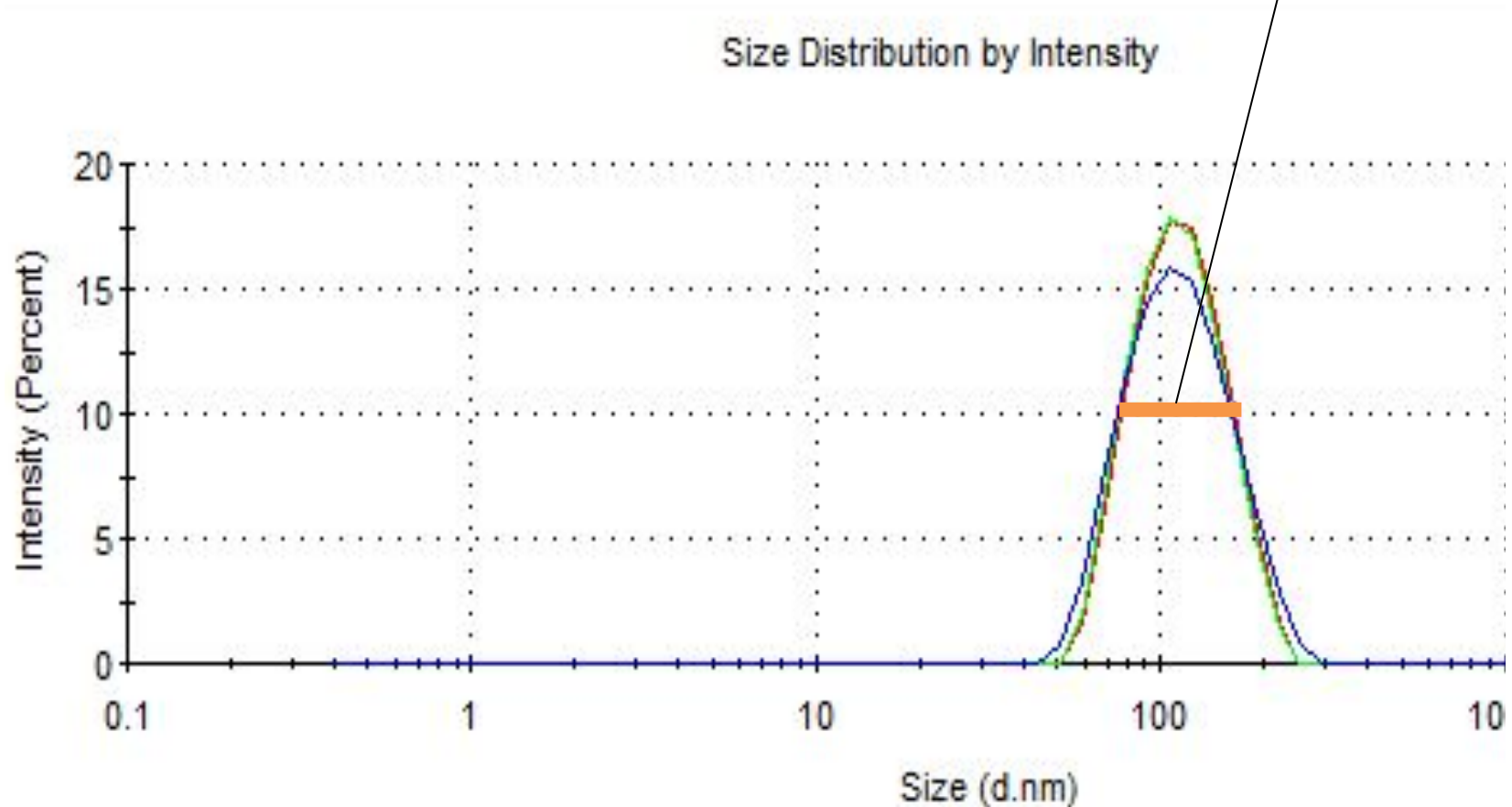


[wikipedia.org]

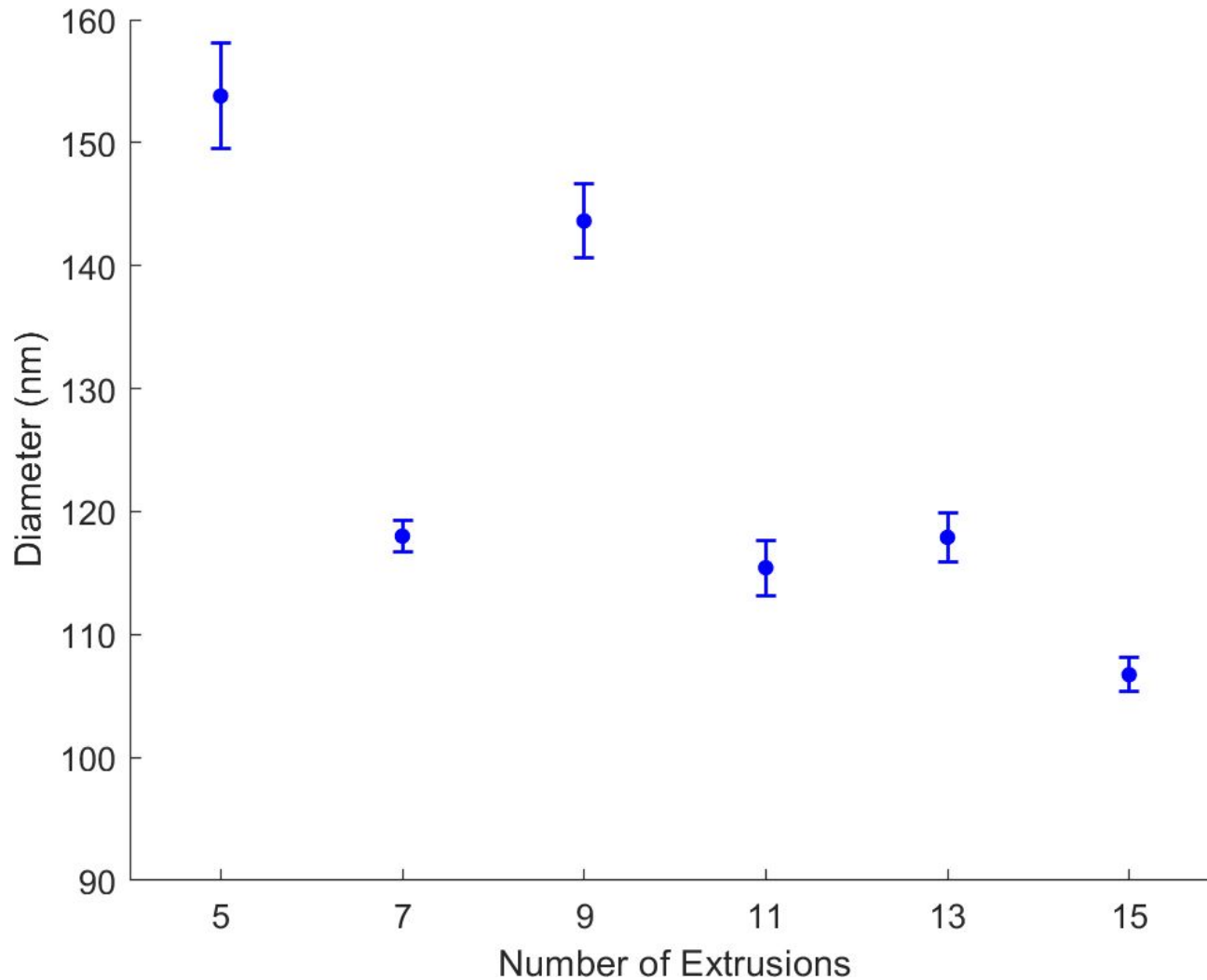


Dynamic Light Scattering (DLS) is used to determine the average particle diameter and Polydispersity Index (PDI).

PDI: width of distribution

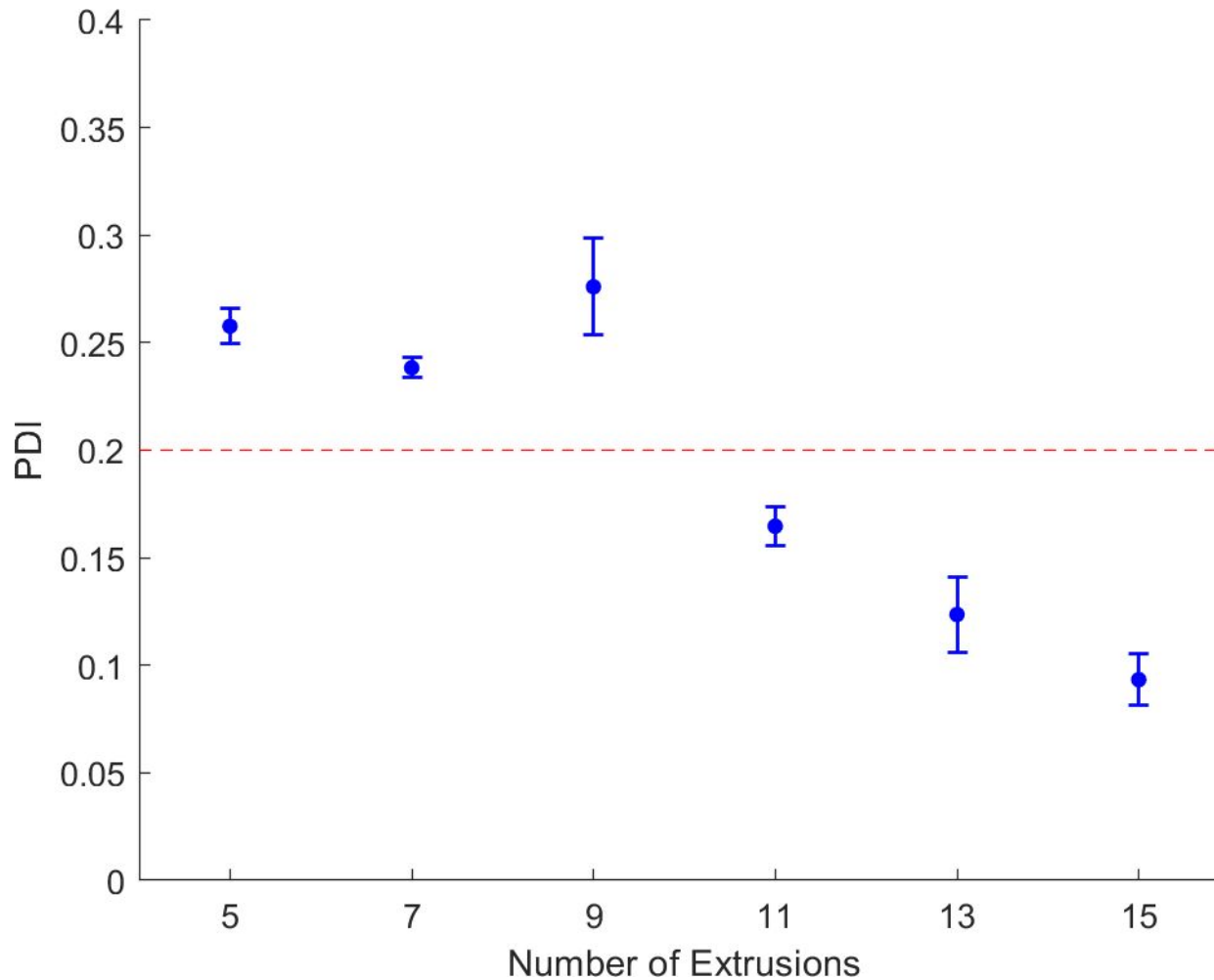


As the number of extrusions is increased, the nanoparticle size decreases.



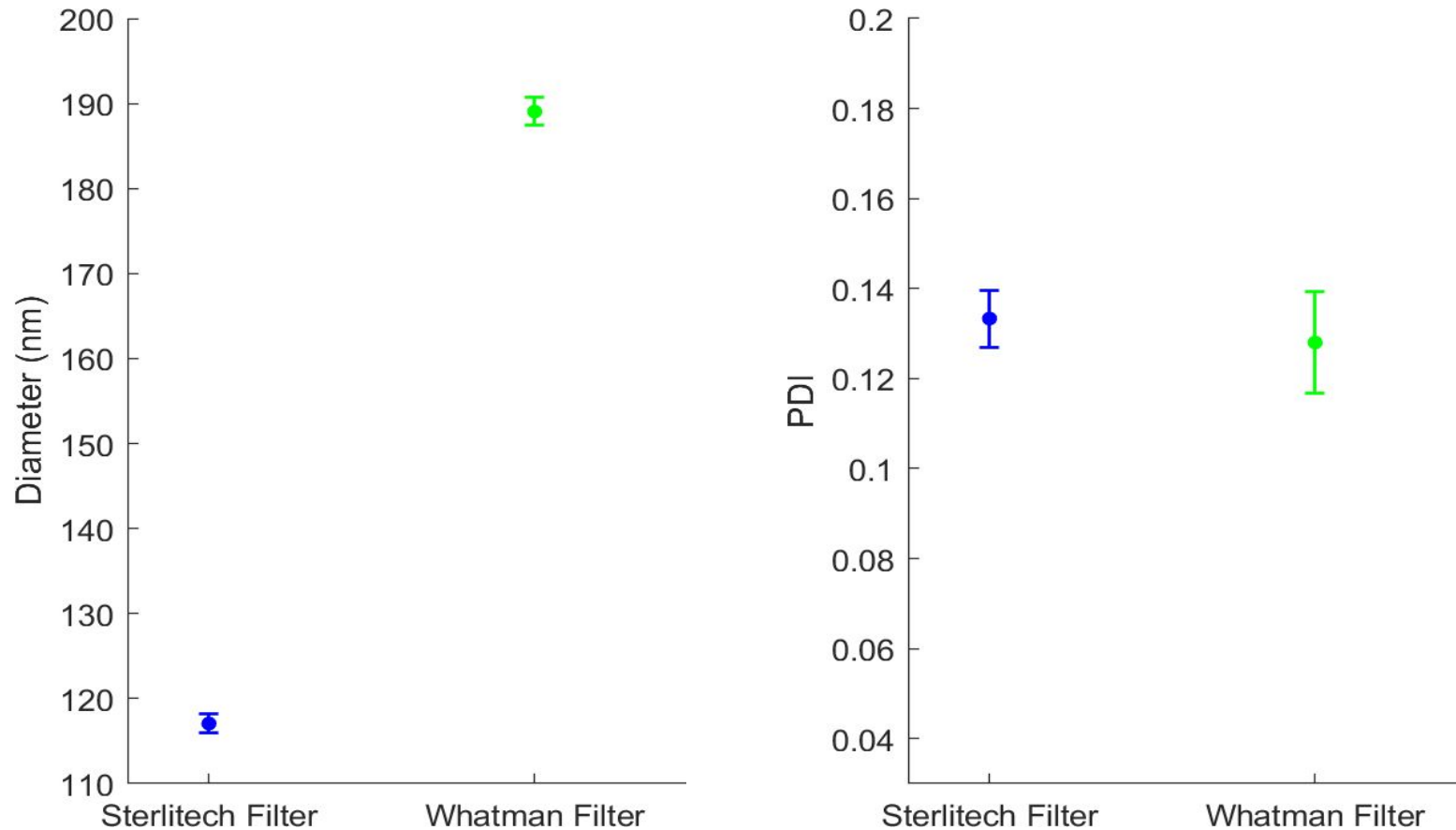
Extrusion	Diameter (nm)
5	154 ± 4.30
7	118 ± 1.31
9	144 ± 3.01
11	115 ± 2.25
13	118 ± 2.01
15	107 ± 1.38

As the number of extrusion increases, the PDI decreases.



Extrusion	PDI
5	0.258 ± 0.008
7	0.227 ± 0.005
9	0.276 ± 0.023
11	0.165 ± 0.011
13	0.124 ± 0.018
15	0.094 ± 0.059

Membrane filters from different brands can also have an impact on experimental outcome.



Filter	Particle Diameter	PDI
Sterlitech	117 ± 1.14	0.133 ± 0.006
Whatman	189 ± 1.63	0.128 ± 0.011

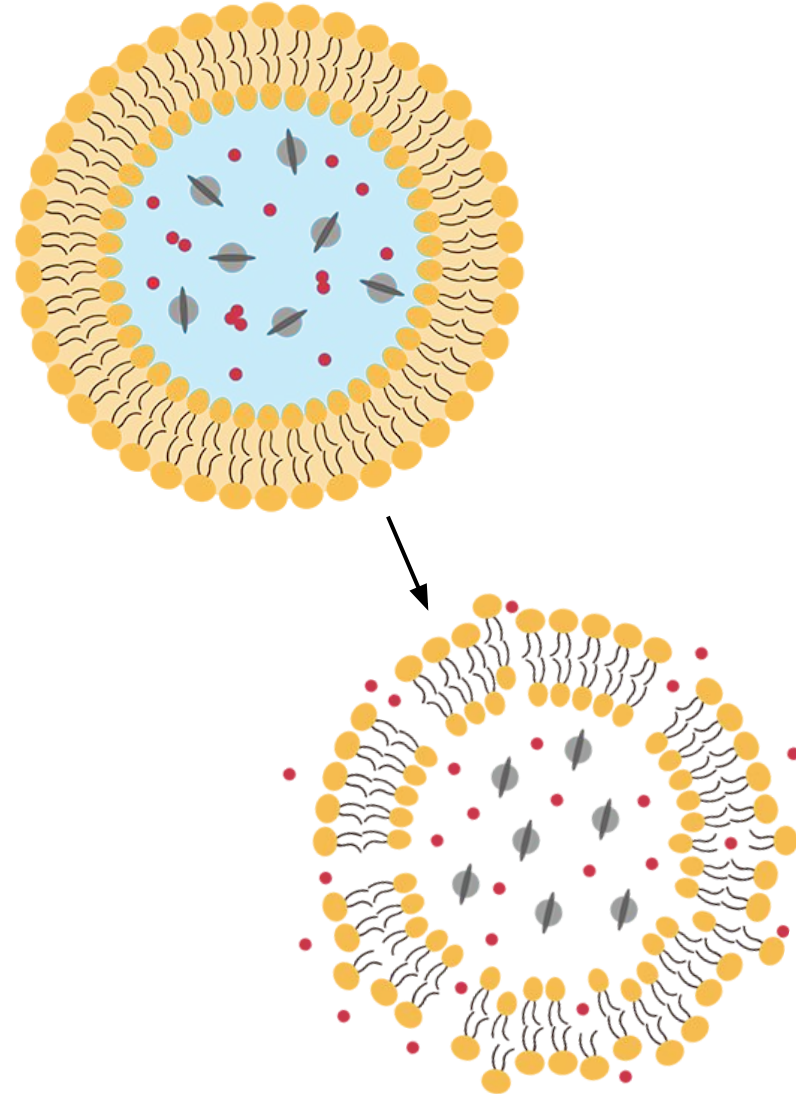
In conclusion, more extrusions results in more desirable liposome nanoparticles by industry standard.

15 extrusions resulted in average particle diameter closest to the targeted 100 nm with the highest uniformity.

Sterlitech filters produced a smaller average particle size.

Whatman filters produced slightly more uniform particles.

In the future, trials for each extrusion should be repeated for more accurate results and extrusions beyond 15 should also be tested.



Questions?